## Q. Implement following:

A. WAP to print twin primes less than 1000. If two consecutive odd numbers are both prime then they are known as twin prime.

B. Write a program to implement these formulae of permutations and combinations. Number of permutations of n objects taken r at a time: p(n, r) = n! / (n-r)!. Number of combinations of n objects taken r at a time is: c(n, r) = n! / (r!*(n-r)!) = p(n,r) / r!

## 1. Code for experiment/practical:

**A.**

```
def is_prime(num):
    if num < 2:
        return False
    for i in range(2, int(num ** 0.5) + 1):
        if num % i == 0:
            return False
    return True

# Loop through odd numbers and check for twin primes
for num in range(3, 1000, 2):  # Only check odd numbers starting from 3
    if is_prime(num) and is_prime(num + 2):  # Check if both numbers are prime
        print(f"({num}, {num + 2})")
```

**B.** import math

```
# Function to calculate permutations: P(n, r) = n! / (n-r)!
def permutations(n, r):
    return math.factorial(n) // math.factorial(n - r)

# Function to calculate combinations: C(n, r) = n! / (r! * (n-r)!)
def combinations(n, r):
    return permutations(n, r) // math.factorial(r)

# Input from the user
n = int(input("Enter the value of n (total number of objects): "))
r = int(input("Enter the value of r (number of objects taken at a time): "))

# Validate input
if r > n:
    print("r should be less than or equal to n.")
else:
    # Calculate permutations and combinations
    p = permutations(n, r)
    c = combinations(n, r)

    # Output the results
    print(f"Permutations P({n}, {r}) = {p}")
        print(f"Combinations C({n}, {r}) = {c}")
```

## 2.Result/Output/Writing Summary:

**<u>Output for Twin prime number</u>**

```
(599, 601)
(617, 619)
(641, 643)
(659, 661)
(809, 811)
(821, 823)
(827, 829)
(857, 859)
(881, 883)

=== Code Execution Successful ===
```

## Output for Combination and Permutation

```
Output                                                    Clear
Enter the value of n (total number of objects): 5
Enter the value of r (number of objects taken at a time): 2
Permutations P(5, 2) = 20
Combinations C(5, 2) = 10

=== Code Execution Successful ===
```

## Learning outcomes (What I have learnt):

1. Learned about Twin prime number.
2. Learned about how range function works.
3. Learned about for loop.

**2. Learning outcomes (What I have learnt):**

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